

## Why a Meshtastic Link Is Not an Application Protocol

Design and field behaviour of the Tricorder SUBSPACE wire format for phone-to-phone exchange over a LoRa mesh card.

Project Tricorder implementation notes. Application build 0.6.111, versionCode 146. 5 October 2026.

### Abstract

Pairing two phones to Meshtastic radios, and confirming that those radios can hear one another, does not cause the phones' applications to exchange structured information. Meshtastic supplies a routed byte pipe: a LoRa physical layer, a mesh forwarder, and a Bluetooth Low Energy session that carries firmware protobufs between one phone and one card. It does not define what a detection, a push-to-talk utterance, or a position report means. This report states that gap as an engineering problem, lists the constraints that a phone application must solve itself, compares the obvious ways of filling the gap, and specifies the text wire format used by the Tricorder field build. The format, called SUBSPACE in the application, rides inside ordinary Meshtastic text payloads. It defines four message classes (push-to-talk, aloha position, framed detection, and a legacy single-line detection), a fragmentation and reassembly rule sized to the card's text budget, an echo rule so a received event is not offered back onto the mesh, and a host-side rule for sharing the radio's single GATT link with local speech. The specification is written so that a reader can implement or audit the exchange without inspecting the application.

Keywords: LoRa, Meshtastic, application protocol, fragmentation, push-to-talk, position report, GATT.

### 1. The problem

An observer who has used a handheld radio tends to treat "the radios are linked" as synonymous with "the operators can now pass the information the application cares about." That equivalence holds for voice on an analog channel, because the application and the channel are the same thing. It does not hold for a smartphone application sitting above a Meshtastic card.

Meshtastic is firmware for a small LoRa radio. The card routes packets among other cards that share a region, a modem preset, and a channel. The phone does not speak LoRa. It speaks Bluetooth Low Energy to one card, using the Meshtastic GATT service, and it exchanges ToRadio and FromRadio protobufs with that card. A text message is one payload type inside that protobuf. The card may beep when it transmits or receives. The beep is evidence that a packet moved. It is not evidence that a second application understood the packet.

The concrete failure mode is easy to stage. Two Galaxy Fold phones, each bonded to a SenseCAP T1000-E, can show a link in the official Meshtastic application. Close that application, because it holds the only GATT session the card will serve. Open a second application that has never defined a message of its own. Nothing about flight detections, spoken messages, or map positions crosses. The radios are doing their job. The applications have not agreed on a job.

The Tricorder application needed three jobs: a spoken message from one operator to the other, a periodic position and friendly name so each navigation plot could show the other unit, and a rate-

limited forwarding of selected local detections (flight, civic, weapon, human, vehicle, flora, soundscape, motion, silence). None of those jobs is a Meshtastic primitive.

## **2. What the radio actually provides**

The useful contract, from the application's side, is narrow.

The phone may write a text string into a ToRadio protobuf and ask the card to transmit it to a destination node identifier. The destination used here is the broadcast identifier ffffffff, so any card on the channel may deliver it. The card applies its own hop limit, duty cycle, and channel key. The receiving card delivers a FromRadio protobuf to whichever phone currently owns the GATT session. The text, if the protobuf decodes, is an opaque string. The card does not parse vertical bars, does not reassemble multi-packet application messages, and does not know that one string is a position and another is speech.

Several host constraints sit on top of that contract. The official Meshtastic application and the Tricorder application cannot both hold the GATT link. A write from the phone can collide with a notification read, so an unsent local packet can eat the inbound FromRadio that carried the remote packet. The practical text budget used by this build is a line under 200 characters, with a 140-character body, because longer strings are clipped or fail to arrive intact. The card's transmit beep is produced by firmware and is not an application acknowledgement. Speech recognition on Android delivers a growing transcript; if each partial result is treated as a finished message, the far unit speaks the message three times.

## **3. Challenges**

3.1 No shared schema. Two independent processes can exchange bytes and still share no types. A detection in the sending application is a structured record: category, time, place, label, and a local identifier. On the air it has to become a string the other process can recognise without a prior session. If the receiver treats every inbound string as display text, a remote detection is spoken as chatter, or worse, offered back to the mesh as if it had been sensed locally.

3.2 Size. A flight card's spoken line exceeds one Meshtastic text payload. Cutting at a fixed character count without a frame header leaves the receiver with a fragment that ends mid-word and no way to know that more is coming. Sending the fragment as a complete message is what produced the field report of a flight line truncated at the word FROM.

3.3 Ordering, loss, and duplication. LoRa delivery is best-effort. Frames can arrive out of order, a middle frame can be lost, and the same text can be observed twice: once from a decoded protobuf and once from a raw-byte scan of the same notification. A protocol that speaks on every observation will repeat itself. A protocol that waits forever for a missing frame will stay silent, which is the failure seen when reassembly required a perfect set and a bad frame returned without speaking.

3.4 Echo. If a received detection is inserted into the same offer queue that local sensors use, the second phone will transmit it back. The first phone will receive it and transmit it again. Civic messages were observed bouncing for this reason. Echo suppression cannot be "ignore my own node," because the

application does not reliably learn a stable node number from the card. It has to be a property of the message class: remote frames are terminal.

3.5 Contention on one link. The phone is both a sensor and a radio terminal. Local text-to-speech, a local detection, and an inbound read share one GATT session and one speech queue. An early build dropped a human detection while speech was active, and dropped a remote utterance when a local detection overwrote a single pending slot. The protocol has to say which side yields.

3.6 Presence without a directory. Each plot knew its own GPS fix. Neither plot knew the other unit's name or position, because a fix computed on one phone is not a packet. A position report has to be small enough to send often, named so the plot can label it, and distinct from the sender's own fix so a unit does not plot itself as a remote.

#### **4. Candidate solutions**

Four approaches were available. They are not equivalent.

Rely on the Meshtastic application. The official application already displays text and positions for nodes that the firmware knows. It does not classify a Tricorder detection, does not speak a remote detection as a remote detection, and cannot run beside this application because it owns the GATT session. It solves the radio problem and leaves the application problem untouched.

Send one unstructured sentence. This is enough for a short push-to-talk line and was the first working path. It fails as soon as the sentence is longer than the payload, and it fails the echo test, because the receiver cannot tell a detection from a sentence it should offer again.

Use a binary protobuf of our own inside the text field, or a second BLE characteristic. A private binary encoding is denser, but the card still carries it as an opaque payload, and a field failure becomes unreadable on the radio log. A second characteristic is not offered by the card. The firmware already has one text pipe. Inventing a side channel would not remove the need for a schema.

Define a small text schema on the existing pipe. This is the approach taken. The cost is framing overhead and a character budget. The gain is that a received payload can be classified by its first token, logged as text, fragmented on a stated boundary, and refused as a candidate for retransmission. That is an application protocol. The radio remains a carrier.

#### **5. The system that was built**

The implementation has four layers, only one of which is new.

The sensor layer is unchanged: audio classification, flight and civic watchers, and the local field log. A detection is offered to SUBSPACE only if its category is enabled. Offering stores one pending record per category. A token bucket, default one per minute and adjustable from one to five, decides when a pending record may be transmitted. Priority is weapon, civic, flight, human, vehicle, flora, motion, soundscape, silence.

The wire layer turns that record into one or more TF frames, or sends a PTT or AH line directly. Frames are handed to a drip queue that waits 700 ms between writes, so the card can drain FromRadio between them. The destination is the channel broadcast.

The link layer is the existing MeshLink GATT client. It writes ToRadio, reads FromRadio, and, if protobuf extraction yields no text, scans the raw notification for the tokens PTT|, AH|, TF|, and TC|. A scan hit is what made push-to-talk and aloha audible when the decoder missed the body. The scan is a fallback, not a second protocol.

The presentation layer speaks or plots only after the wire layer has accepted a message. Push-to-talk is spoken once, as a remote utterance. A detection is spoken only after its frames reassemble, or after a 2.5 second partial-set timeout. An aloha updates the navigation plot and is not spoken as a detection. The plot labels the remote unit with name, time, and position, including when the two handsets are within a few metres of each other.

## **6. How the design meets the challenges**

No shared schema: a leading token selects the handler. Unknown text is not offered to SUBSPACE.

Payload size: detection bodies are cut into 140-character chunks. Each chunk carries index and count.

Loss: a single-frame message speaks immediately. A partial set speaks after 2.5 seconds rather than never.

Duplication: push-to-talk sends on the final recognition result only. A growing repeat replaces the shorter utterance.

Echo: received TF and AH are terminal. They update speech or the plot and are not re-offered.

GATT contention: writes and FromRadio reads are serialised. A send is followed by a drain.

Speech contention: remote speech is queued. A local detection no longer overwrites the only pending utterance.

Presence: AH is sent on a timer once the radio is linked and a GPS fix exists, not only when a detection fires.

Self-plot: an aloha is ignored only when both the friendly name and the radio-address tail match the local unit.

Decoder miss: raw notification bytes are scanned for the same tokens the decoder would have returned.

Each of those is a failure observed in the field, not a hypothetical.

## **7. Wire format**

This section is the normative note for build 0.6.111. Octets are ASCII. The field separator is the vertical bar. A sender must not put a raw vertical bar inside a field; the implementation replaces it with a slash. The payload is the Meshtastic text string. The radio header, hop count, and channel key are out of scope.

7.1 Common rules. A receiver classifies a payload by its prefix. The first matching rule wins. Payloads that match none of the prefixes are not SUBSPACE messages and must not be offered for retransmission. Comparison of the prefix is case-sensitive. Names are trimmed and limited to 16

characters on the aloha path. Times are local HH:mm, 24-hour, from the sending phone. Coordinates are decimal degrees.

## 7.2 PTT, push-to-talk.

text

PTT|<from>|<text>

from is the sender's friendly name. text is the finished utterance. The sender must emit this line once, when the recogniser marks the result final, not on each partial transcript. The receiver speaks "From . " and does not offer the line to the detection queue. If a second PTT from the same from arrives within 8 seconds and its text extends the previous text, the receiver stops the shorter utterance and speaks the longer one. That rule absorbs a partial that escaped the sender.

## 7.3 AH, aloha, the presence report.

text

AH|<name>|<tail>|<lat>|<lon>|<hhmm>

name is the friendly name. tail is the last four hexadecimal characters of the bonded radio address, or the last four characters of the local guid if the address is not yet known. lat and lon are decimal degrees to five places. hhmm is the send time. The line is one frame. The sender arms the timer when the radio session starts, attempts the first send after about 8 seconds, and retries every 20 seconds until both a link and a GPS fix exist. After a successful send the interval is the operator setting: 3 minutes or 1 minute. The receiver plots the point, stores name and tail, and does not speak it as a detection. It discards the report only when name and tail both match the local unit. A report from the same name with a different tail is a different unit.

## 7.4 TF, framed detection.

text

TF|<id>|<i>/<n>|<cat>|<chunk>

id is four characters taken from the sender guid. i and n are decimal, i starting at 1. cat is one of weapon, civic, flight, human, vehicle, flora, motion, soundscape, silence. chunk is at most 140 characters. The logical body is the concatenation of chunks in index order. The body begins with the detection time, a space, the place as lat,lon to three places or the token na, a space, and the label. The label is capped at 480 characters before framing, so n is at most 4 for the current cap. A sender drips frames 700 ms apart. A receiver that has i = 1 and n = 1 speaks immediately. A receiver that has a partial set speaks the concatenation of the frames it holds after 2.5 seconds. It does not wait indefinitely. It does not offer the reassembled text to the local send queue.

Example, a two-frame flight report:

text

TF|A1B2|1/2|flight|09:14 35.227, -80.843 Overhead flight from east

TF|A1B2|2/2|flight| bound, registration N12345

7.5 TC, single-line detection.

text

TC|<guid>|<cat>|<rest>

TC is the earlier single-line form. A receiver that still sees it must treat it as a finished remote detection, not as local context. Wrapping a TC line as unstructured context was the fault that kept flight and civic messages out of the parser. New senders use TF.

7.6 Receiver state machine. On each accepted text the receiver applies the first matching rule.

If the text starts with AH|, parse five fields after the token. On success, update the peer plot and stop. If it starts with PTT|, split into from and text on the first two bars, speak, and stop. If it starts with TF|, insert the chunk into the buffer keyed by id and category. If the set is complete, or the frame is 1/1, speak and clear. Otherwise arm the 2.5 second timer. If it starts with TC|, speak as a remote detection and stop. Otherwise ignore the text for SUBSPACE purposes.

Speaking a remote detection uses the form "Remote tricorder detected . ". Speaking must not enqueue a new offer. The field log may record the line. The send queue may not.

7.7 Sender scheduling. Detection sends and presence sends are separate schedulers. The detection scheduler will not transmit unless the operator has enabled LoRa and the category. It holds at most one pending record per category, so a newer observation of the same category replaces an unsent older one. It spends one token per logical detection, not per frame. The presence scheduler does not spend detection tokens. Both share the same drip queue onto the card, so a detection burst cannot overlap an aloha write without the 700 ms gap.

## 8. What this protocol is not

It is not a replacement for Meshtastic. Node identity, routing, encryption, and the region plan remain firmware concerns. It is not a reliable transport. A lost frame is spoken as a partial or not at all. It is not addressed to a unit: the antenna list on the handset selects a peer for the operator, but the PTT line is still broadcast, and every Tricorder on the channel will speak it. Private delivery would require a destination node number the application does not yet learn reliably from the card. It is not authenticated. A third card on the channel can forge an AH or a PTT line. Channel encryption, if configured in the firmware, is the only confidentiality, and it is not message authentication of the application fields.

Those limits are why the result can surprise someone who has only seen the cards beep at each other. The beep confirms the carrier. The vertical-bar grammar confirms the application. One without the other is a working radio and a silent program, which is the state the field units were in before this format existed.

## 9. Conclusion

Linking two phones through Meshtastic cards establishes a routed, best-effort text pipe between whichever phone currently holds each card's GATT session. It does not establish a detection protocol, a position protocol, or a push-to-talk protocol. Those are application agreements. The Tricorder

agreement is a four-token text format, a 140-character frame, a short reassembly timer, a rule that received frames are not re-offered, and a scheduler that separates presence from detections. The format is small because the pipe is small. The format is explicit because the pipe is not.

This note records an implementation. It is not a claim that Meshtastic should have anticipated this application, and it is not a standard offered to other projects. A second implementation can interoperate by following section 7.